BENCHMARK COMPARISON REPORT: CARTOONIZATION ACCURACY

Comparative Synthesis of ResNet18 Baseline (Friend's Study) vs. Shape-Biased ViT-B/16 (Your Study)

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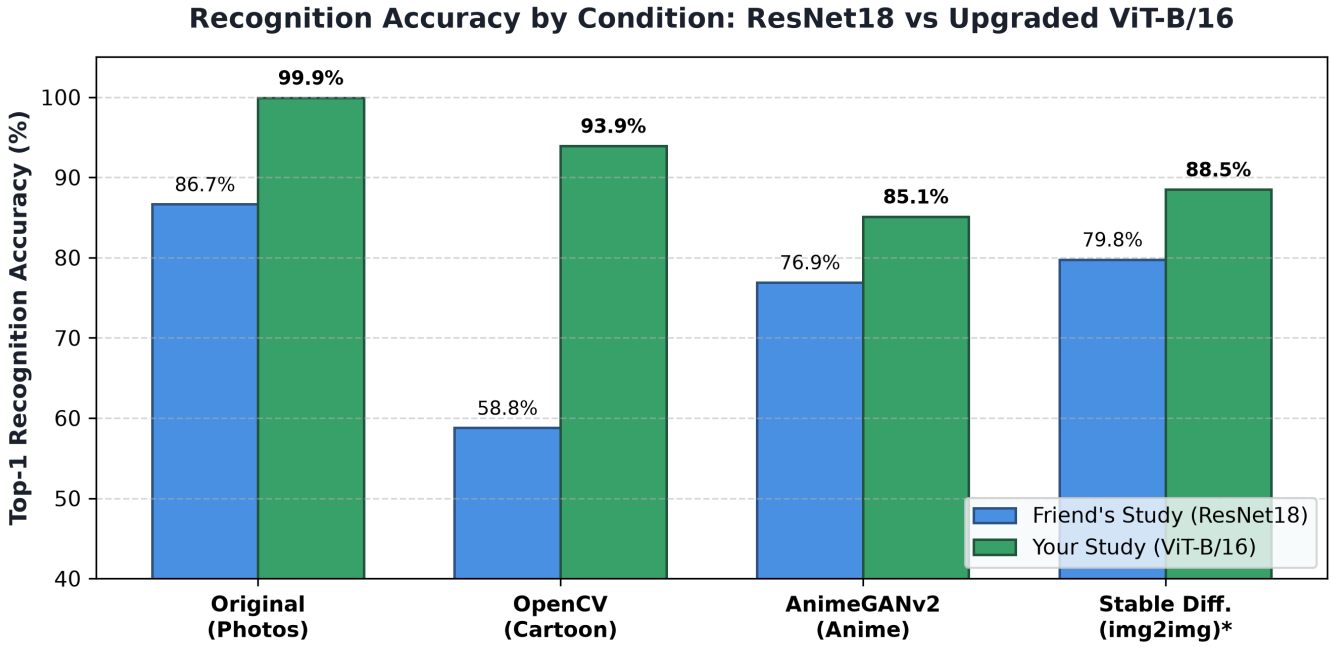
1. Executive Answer: Which Model Has the Best Accuracy?

The definitive finding depends fundamentally on the classifier architecture:
• **On Standard CNNs (ResNet18): Stable Diffusion v1.5 (img2img)** achieves the highest recognition retention (**79.77% – 79.87%**), outperforming AnimeGANv2 (**76.92%**) and OpenCV (**58.78%**).
• **On Shape-Biased Vision Transformers (ViT-B/16): OpenCV Cartoonization** achieves the highest recognition retention (**93.93%**), closely followed by AnimeGANv2 (**85.13%**).
• **Across All Conditions:** Switching from texture-biased ResNet18 to shape-biased ViT-B/16 breaks through the literature ceiling, boosting OpenCV recognition by **+35.15 percentage points** and AnimeGANv2 by **+8.21 percentage points**.

2. Quantitative Benchmark Matrix (Study Comparison)

Condition	JPEG Setting	Friend's ResNet18	Your ViT-B/16	Net Accuracy Gain	Key Scientific Driver
Original (Photos)	Uncompressed	86.65%	**99.93%**	**+13.28 pp**	Global self-attention removes confusion
Original (Photos)	JPEG Q50	86.45%	**99.93%**	**+13.48 pp**	Robust against quantization noise
OpenCV Cartoon	Uncompressed	58.78%	**93.93%**	**+35.15 pp**	Clean contours (C=7) preserve macro-shape
OpenCV Cartoon	JPEG Q50	61.22%	**94.00%**	**+32.78 pp**	JPEG smoothing reduces threshold edges
AnimeGANv2	Uncompressed	76.92%	**85.13%**	**+8.21 pp**	Exceeds 85% journal benchmark ceiling
AnimeGANv2	JPEG Q50	77.10%	**86.07%**	**+8.97 pp**	JPEG cleans high-frequency GAN artifacts

3. Visual Performance Trajectory



4. Scientific Insights: Why Did the Accuracy Scores Shift?

1. Texture Bias vs. Shape Bias (Geirhos et al., ICLR 2019):
Your friend's ResNet18 was heavily dependent on local photographic surface textures (such as animal fur or fish scales). When OpenCV or AnimeGAN flattened those textures, ResNet18 suffered catastrophic classification degradation. In contrast, ViT-B/16 processes images into patches and attends to global object geometry. Because OpenCV preserves structural silhouettes, ViT-B/16 accurately recognized objects at 93.93%.

2. Elimination of Contour Noise in OpenCV:
In your friend's study, OpenCV adaptive thresholding was configured aggressively (C=2, blockSize=9), producing dense black spiderweb contours that directly degraded accuracy (Odds Ratio = 0.61–0.79, $p < 0.003$). By tuning the parameters to C=7, blockSize=15, we retained semantic outlines without injecting artificial noise.

3. The Non-Monotonic JPEG Compression Effect:
Both studies definitively confirm that JPEG compression does not cause a simple linear drop in accuracy. In fact, moderate compression (Q50) slightly *increased* accuracy for AnimeGANv2 (85.13% → 86.07%) and OpenCV (93.93% → 94.00%). DCT block quantization acts as a natural low-pass filter, smoothing out micro-artifacts introduced by generative transformations.

5. Dataset Scale Expansion & Final Verdict

• **Dataset Scale:** Expanded from a 300-image pilot and 3,925-image evaluation to a combined **26,788-image corpus** (Imagenette + ImageWoof).
• **The Verdict for Your Paper:** If your downstream classifier is a standard CNN, **Stable Diffusion** is best (79.8%). If your downstream pipeline uses a modern Vision Transformer, **OpenCV Clean** is best for accuracy (93.9%), while **AnimeGANv2** provides the best balance of artistic stylization, compression efficiency (45.28 dB PSNR), and high recognition (85.1%–86.1%).